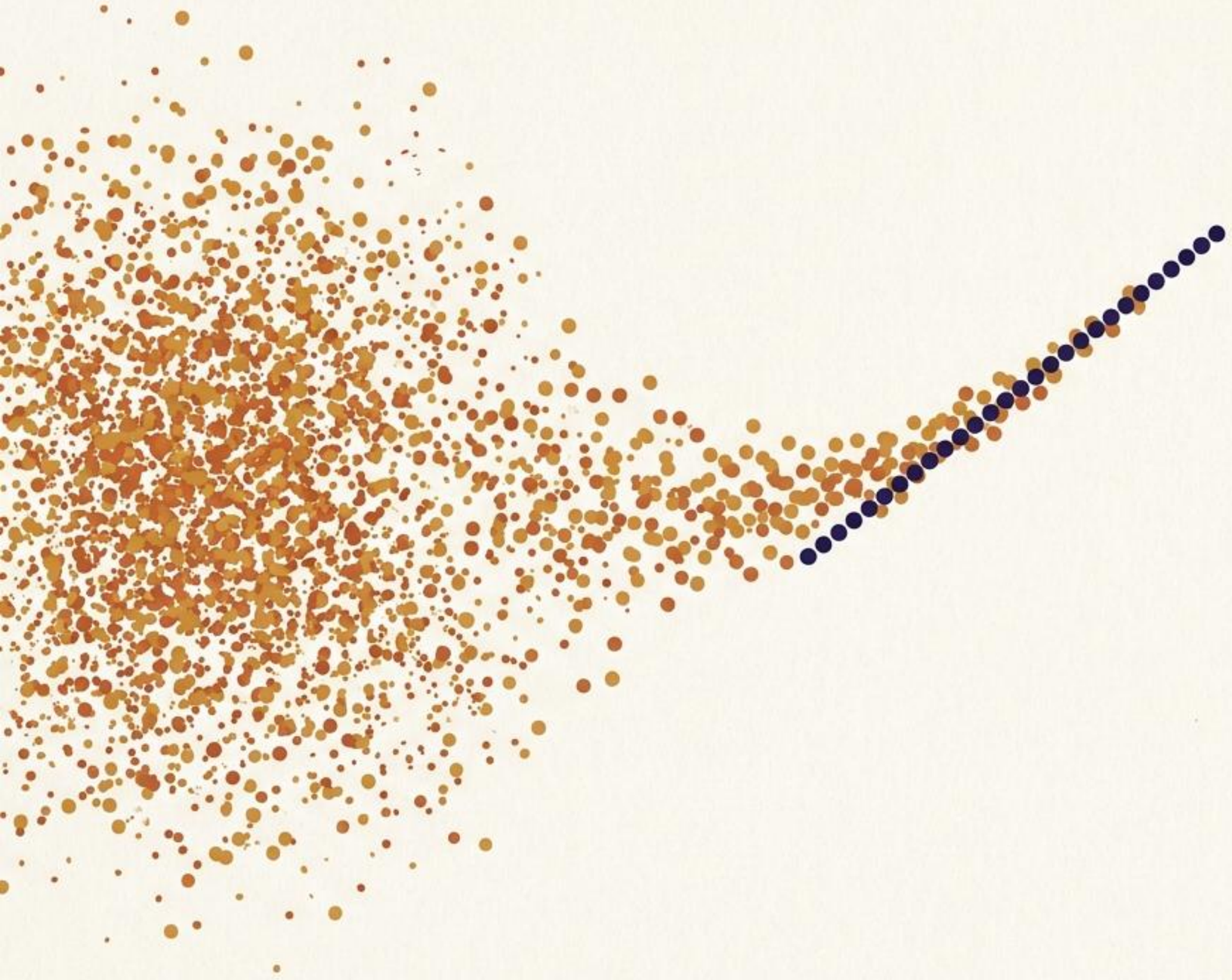
BASIC STATISTICS

Basic Statistics

From data to decisions





Making Sense of Data

A Beginner's Guide to Basic Statistics

Gemini Notebook

The statistical workflow

01

Ask

What do we want to understand?

02

Collect

Who or what should we measure?

03

Describe

What patterns appear in the data?

04

Interpret

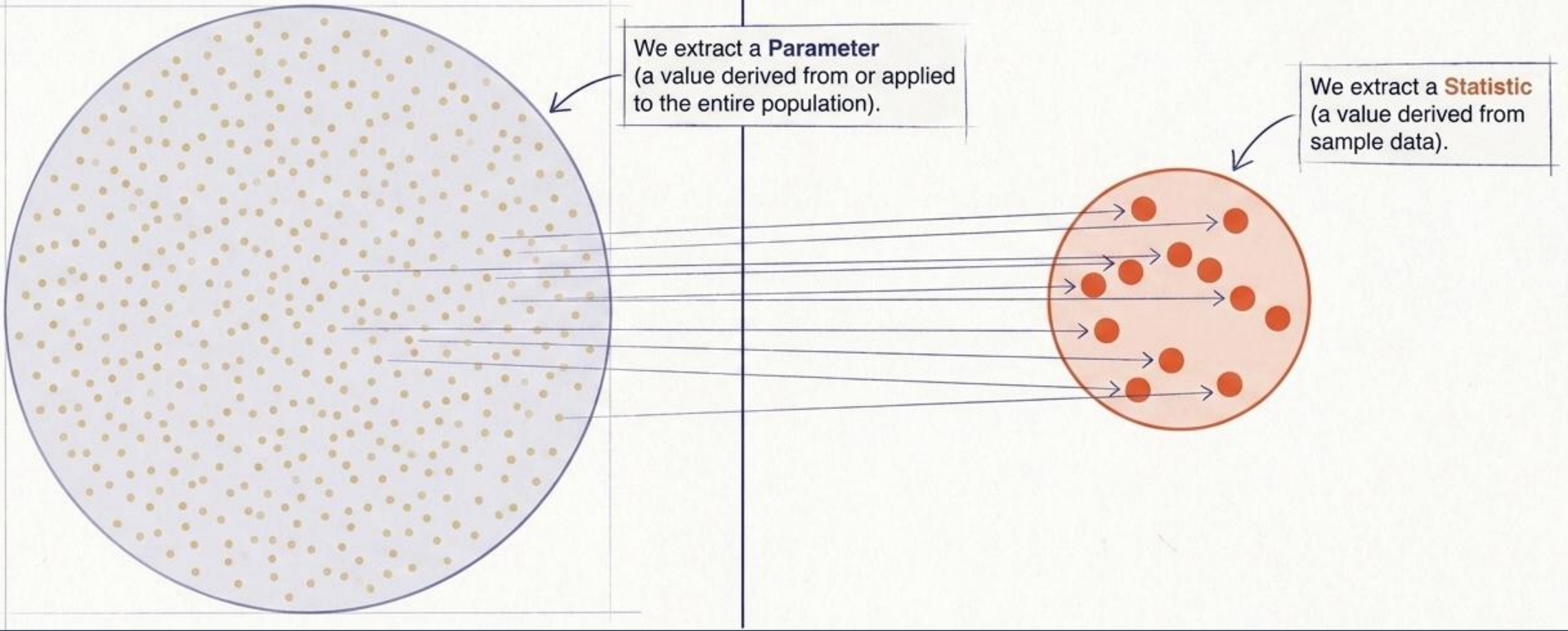
What can we responsibly conclude?

The Ideal

(The whole group of interest)

The Reality

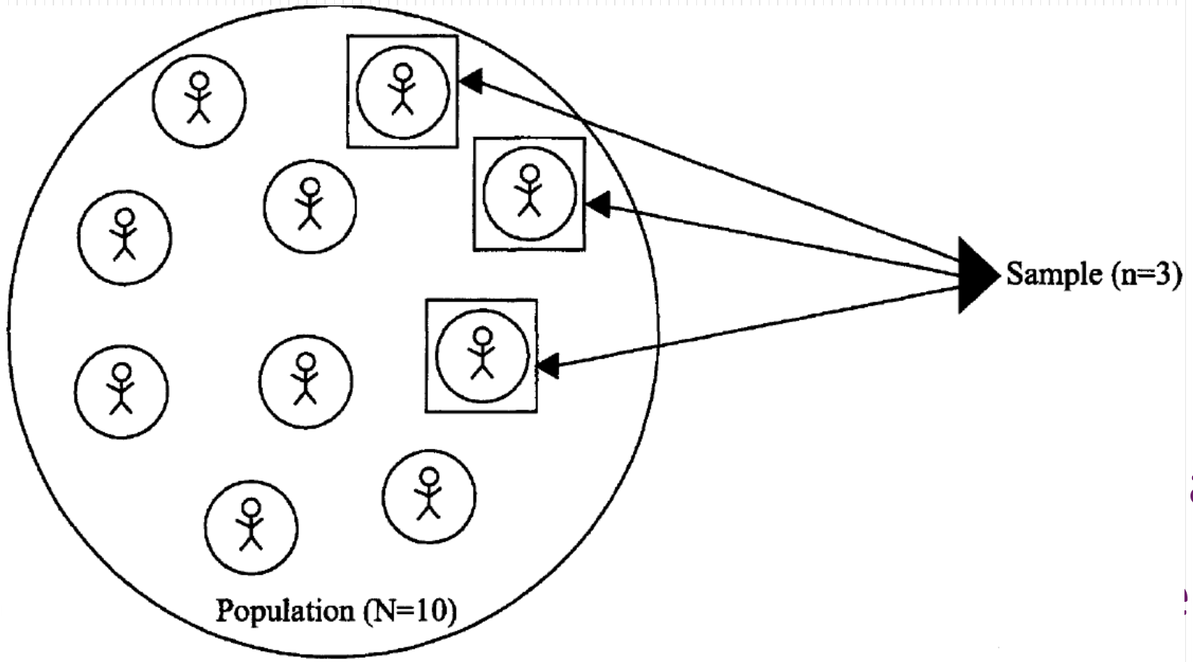
(A subset drawn from the population)



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Populations and Samples

- **Population:** An individual or group that represents all the members of a certain group or category of interest
- **Sample:** A subset drawn from the larger population



ata

Parameters are v d to, population data

Few things to Remember!!!

- Population does not need to be large.
- Samples are not necessarily good representations of the populations



The Grammar of Data

Measurement Scales

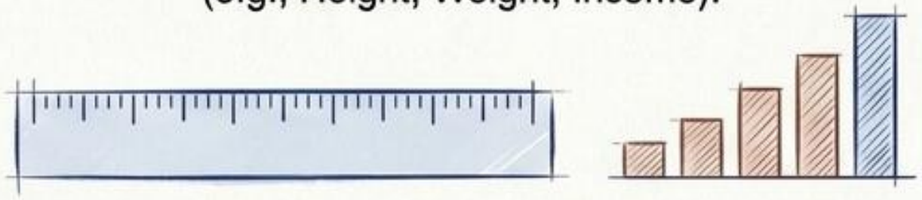
Categorical (Nominal)

Measured in distinct categories with no numerical weight (e.g., Democrat, Republican, Independent).



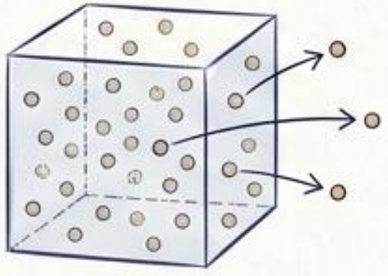
Continuous (Interval/Ratio)

Ordered, numerical, and mathematically meaningful (e.g., Height, Weight, Income).



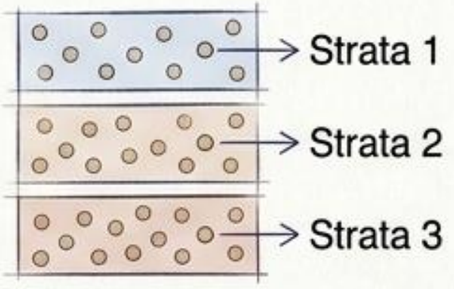
Sampling Methods

Random



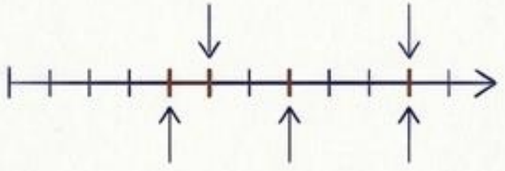
Equal likelihood of selection for all.

Stratified



Population divided into groups, then randomly sampled.

Uniform



Selected at evenly spaced intervals (e.g., every Monday).

Convenience



Based on accessibility.
Warning: highly prone to bias.

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Examples

- **Random sampling:** the likelihood of selection of each member of the population is equal
  - Pick any streamflow value from a population

- **Stratified sampling:** Population is divided into groups, and then a random sampling is used
  - Pick a streamflow value from annual maximum series.

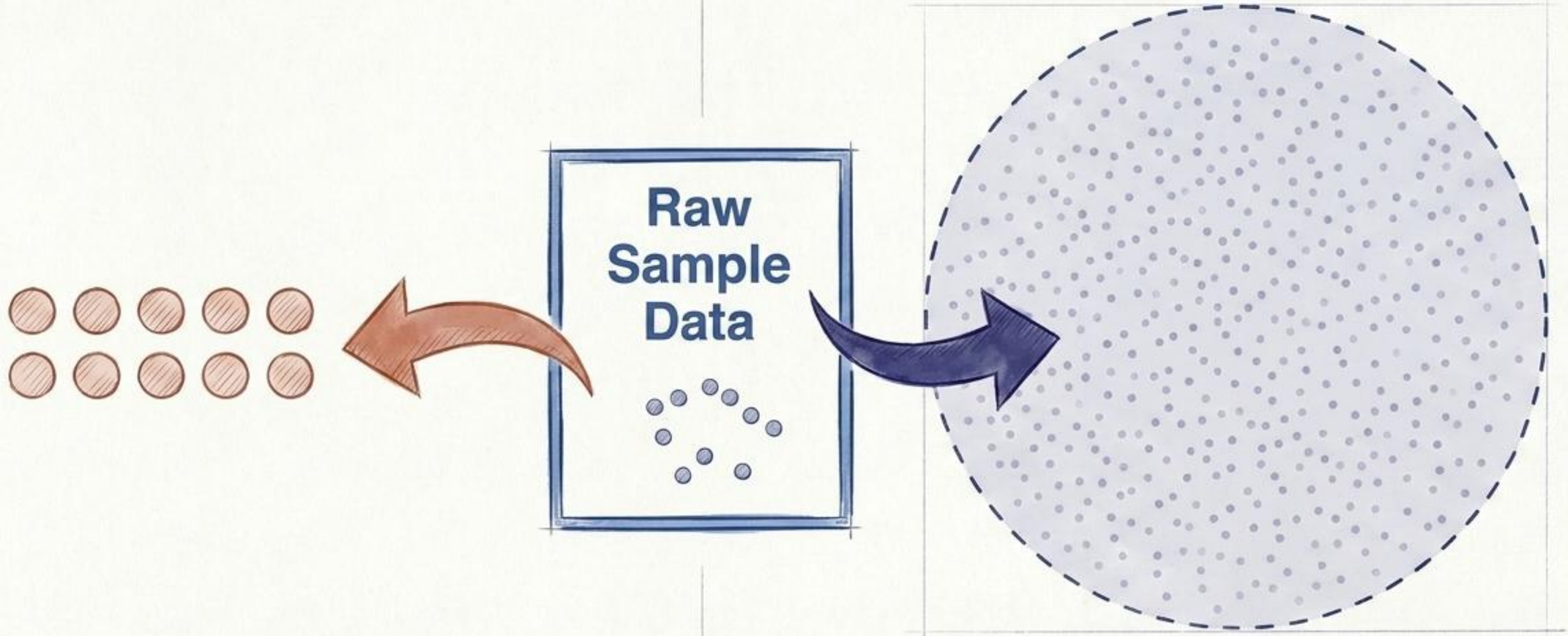
Examples

- **Uniform sampling:** Data are selected such that the points are uniformly far apart in time or space
  - Pick streamflow values measured on Monday midnight

- **Convenience sampling:** Data are collected according to the convenience of experimenter.
  - Pick streamflow during summer

Descriptive Statistics

Inferential Statistics



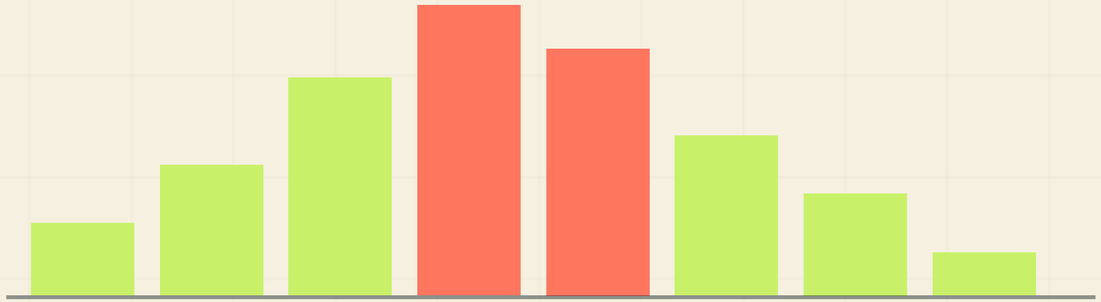
Summarizes the specific characteristics of the collected sample.
(e.g., The average weight of 30 specific adults).

Uses sample data to reach **conclusions** about the larger population. It calculates the probability that an observed effect is due to random sampling error (chance) versus a real phenomenon.

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A distribution shows the shape of data

A frequency table counts how often values occur. A histogram turns those counts into a visible pattern.

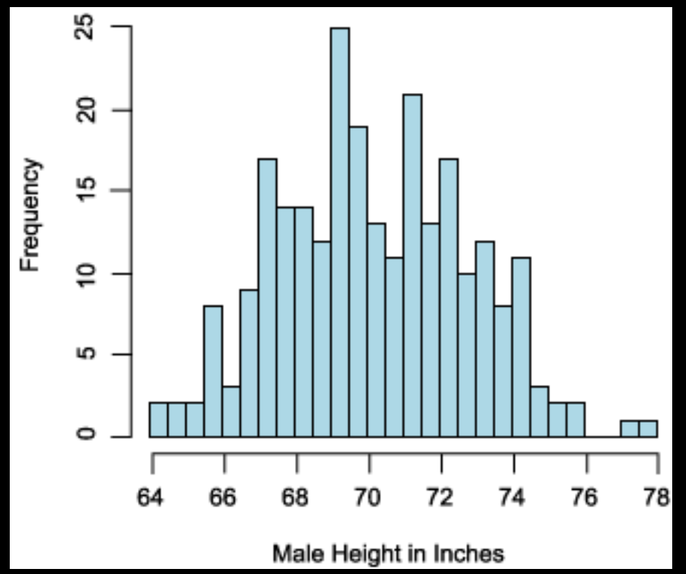


Look for

center
spread
shape
unusual values

Frequency Distribution of a Random Variable

- Some values of a random variable occur more frequently than others
- Can be represented graphically by the ***Frequency distribution*** of the random variable



Three ways to describe the center

Mean

the arithmetic average

sum ÷ count

Median

the middle value when data is ordered

middle position

Mode

the most frequent value

most common

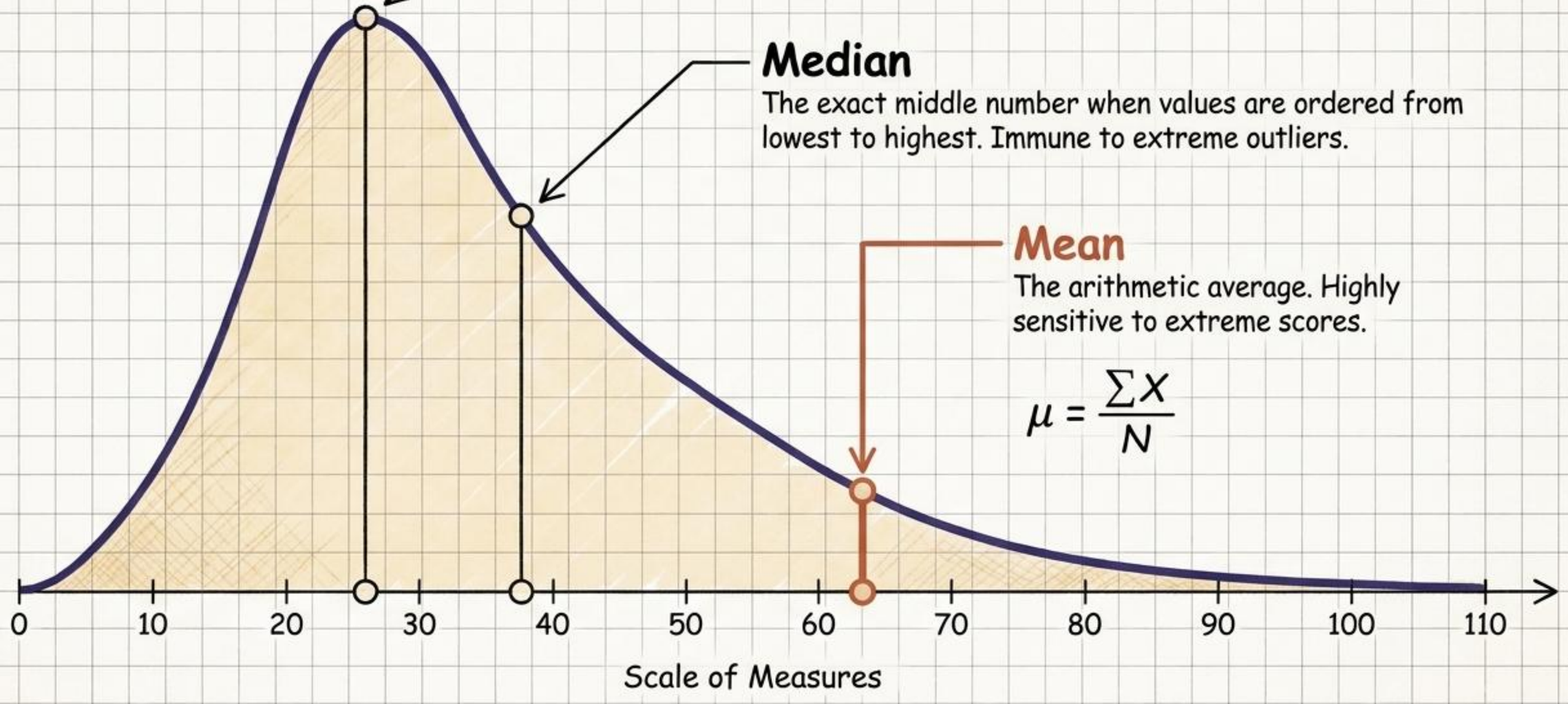
> The best measure depends on the shape of the data.

If the Spatial records are Numbers

- statistical Description

Central Tendency	Mean, Median, Mode
Variation	Range, variance, standard deviation

Measures of Central Tendency & Skewness



Gemini Notebook

Measures of Dispersion: Understanding Variability

1

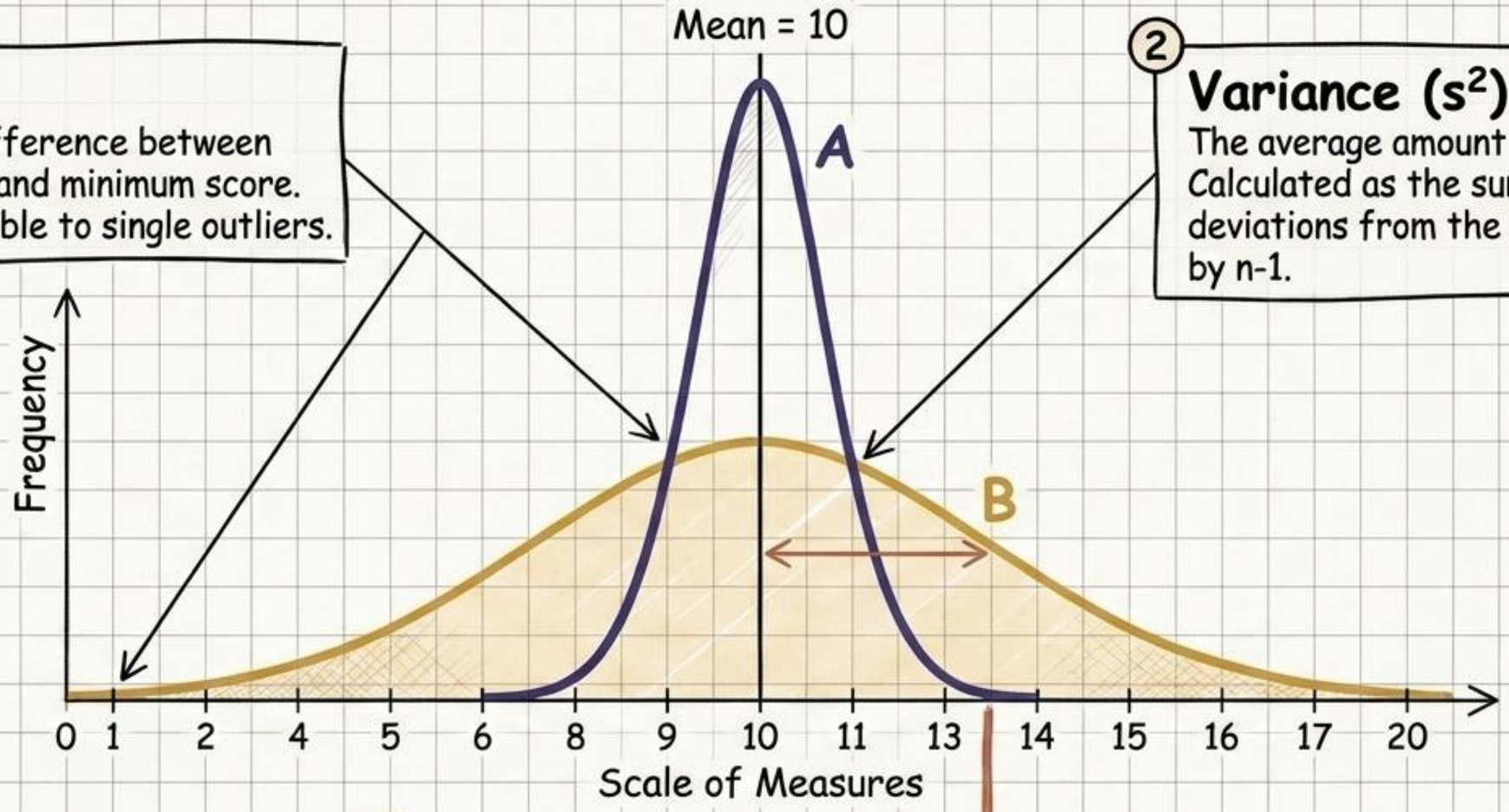
Range

The simple difference between the maximum and minimum score. Highly vulnerable to single outliers.

2

Variance (s^2)

The average amount of dispersion. Calculated as the sum of squared deviations from the mean, divided by n-1.



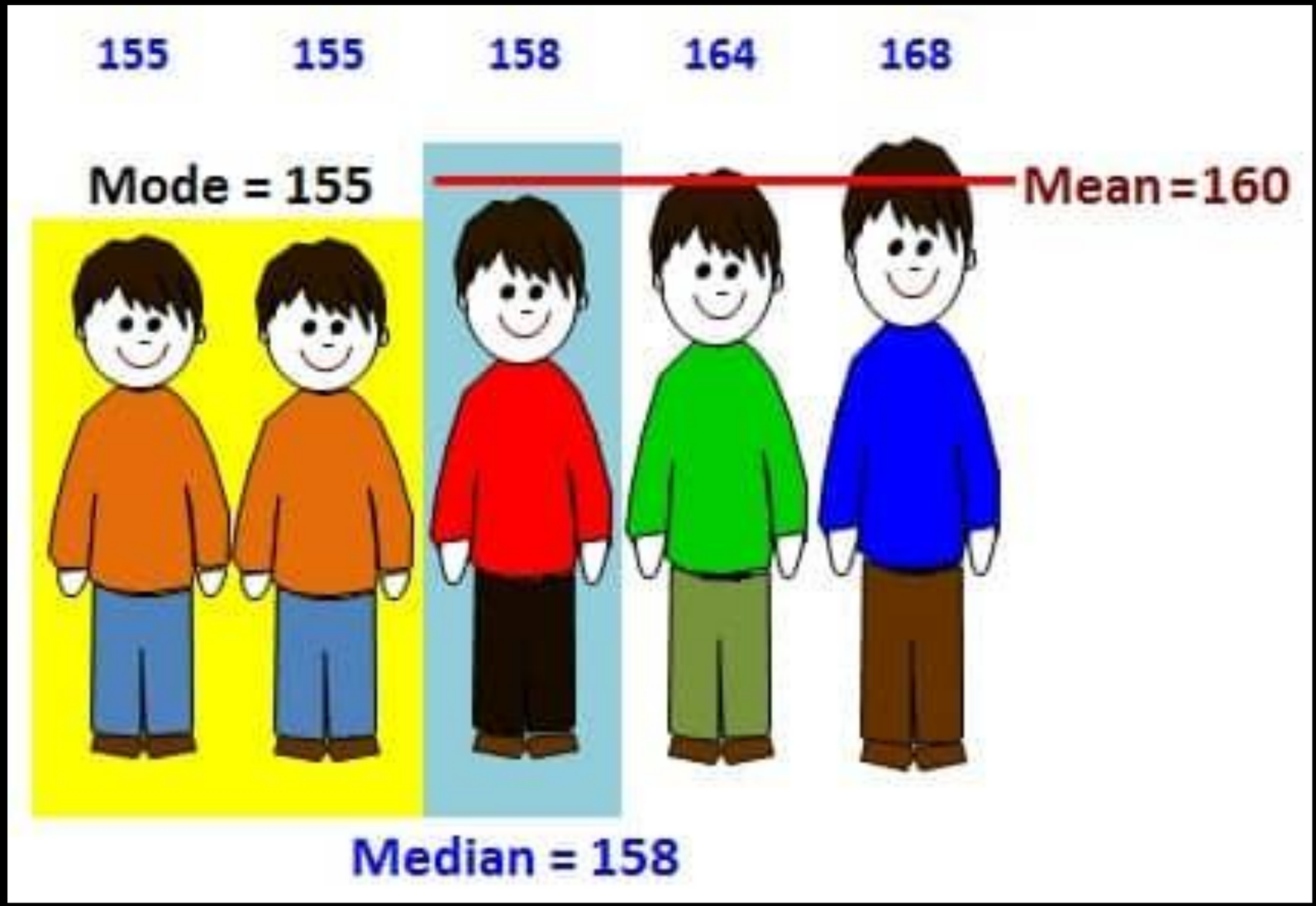
3

Standard Deviation

The square root of variance. The average distance of individual scores from the center.

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Measure of Central Tendency



Slide the outlier in—and watch the mean move

Outlier omitted



Mean = 7.5

Outlier added



Mean = 12.14

6 values

7 values, including 40

Mean

- Statistical average
- Sum of the values for one attribute divided by the number of records

$$\bar{X} = \sum_{i=1}^{n} x_i / n$$



Problem using mean values….

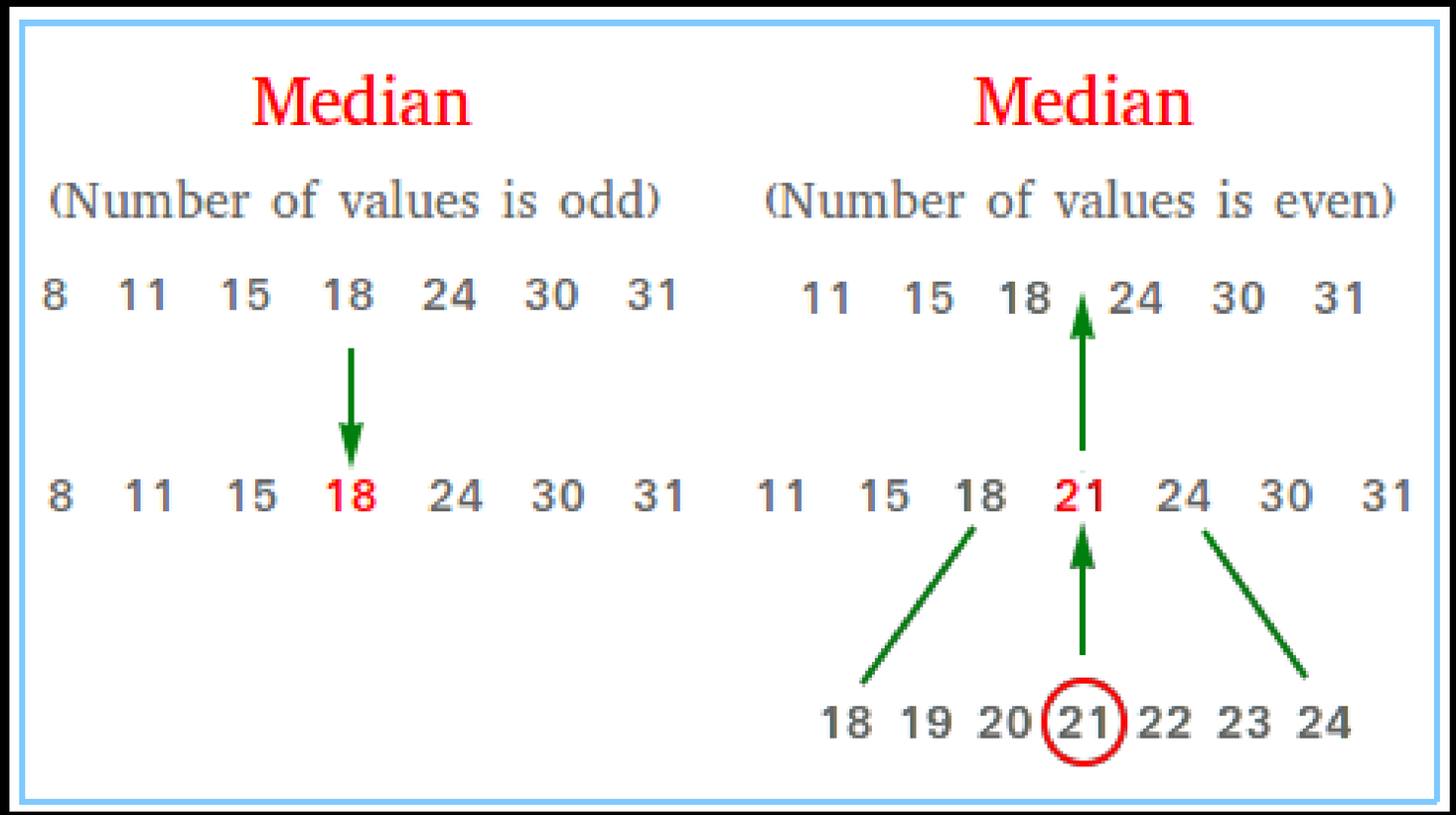
ANY IDEA??

If your head is in the freezer
and your feet are in the oven,
on average, you're comfortable.

Median

- Median is the middle number

- Example



Mode

- The Mode is the most frequent value
- Example

- A mode is the number in the data set that appears the most frequently.
- A data set can have:
  - No mode 1 2 3 5
  - One mode 1 1 2 3 5 one mode: 1
  - Two modes (bimodal) 1 1 1 2 2 3 3 3 5 2 modes: 1, 3
  - More than two modes 1 1 1 2 2 2 3 3 3 5 7 8 9 9 10

Measure of Variability

- Range
- Variance
- Standard deviation
- Covariance

Variance

- Statistical average of the amount of dispersion in a distribution of scores
- The total variance is the sum of each record with its mean subtracted and squared and then divided by the number of records-1

$$\text{Variance} = \frac{\sum (X_i - \bar{X})^2}{n - 1}$$

Standard Deviation

- Average difference from the mean
- The standard deviation is the square root of the variance
- Sum of the mean subtracted from the value for each record, squared, divided by the number of records-1, square rooted.

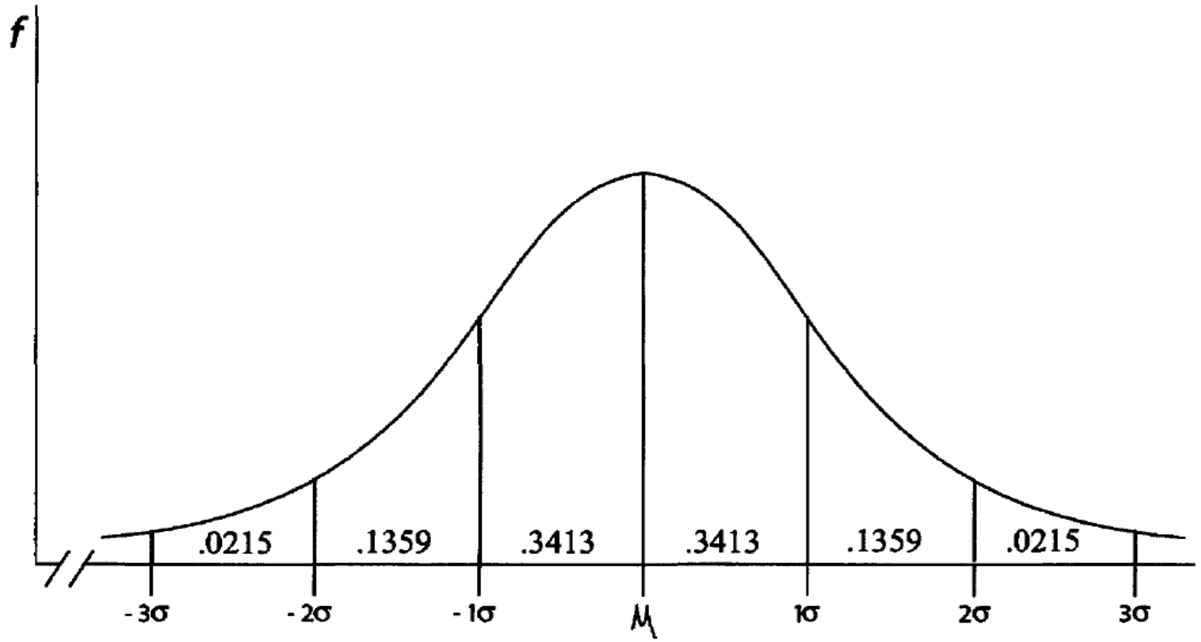
$$\text{st.dev.} = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n - 1}}$$

Range = measure of total spread

Variance and SD = measures of average amount of spread within the distribution.

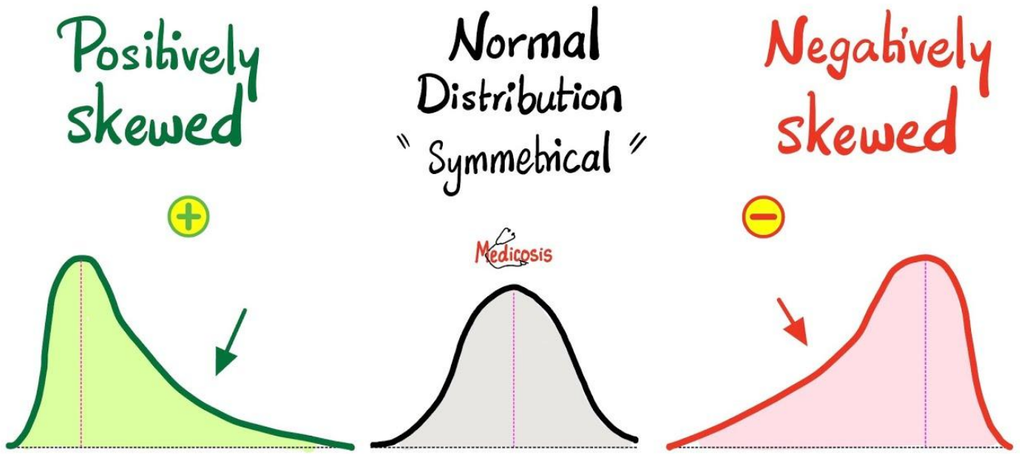
Normal Distribution

- Characteristics of a normal distribution
  - Symmetrical
  - Mean, median and mode are all in the center of the distribution
  - It is asymptotic



Skew

- When a sample of scores is not normally distributed
- **Positively Skewed**
  - Elongated tail at the higher end of the distribution
  - Most of the scores in a distribution occur at the low end, with a few scores at the higher end
- **Negatively skewed**
  - Elongated tail at the lower end of the distribution



Standard Error

- Standard deviation of the **sampling distribution** of some statistic
- Measure of how much random variation we would expect from samples of equal size drawn from the same population
- The standard error of the mean provides the average difference between the expected value (e.g., the population means) and an individual sample mean.
- The standard error of the mean provides a measure of how much error we can expect when we say that a sample mean represents the mean of the larger population.

- The larger the sample, the smaller the standard error.
- The larger the sample standard deviation, the greater the assumed variation of scores in the population, and consequently the larger the standard error of the mean.
- In those instances where you know the population standard deviation, you can use that in the calculation of the standard error of the mean.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

or

$$s_{\bar{x}} = \frac{s}{\sqrt{n}}$$

where σ is the standard deviation for the population
s is the sample estimate of the standard deviation
n is the size of the sample

From Variance to Covariance

- Variance → How does one variable vary?
- Covariance → How do two variables vary together?

Key idea: *Covariance tells us the* ***direction*** *of a relationship.*

Covariance: Do Two Variables Move Together?

- Positive covariance (+): both tend to increase or decrease together.
- Negative covariance (–): one tends to increase when the other decreases.
- Near zero: no clear linear co-movement.

Example

Rainfall ↑ → Streamflow ↑ = Positive covariance

Any Example with Negative and zero covariance???

How Is Covariance Calculated?

For two variables, X and Y:

$$Cov(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

- Think of it in three simple steps:
- Find how far each observation is from its mean.
- Multiply the deviations of X and Y.
- Take their average.

EX

Rainfall (X)	Discharge (Y)	$X - \bar{X}$	$Y - \bar{Y}$	Product
10	20			
20	40			
30	60			

Positive OR Negative covariance?

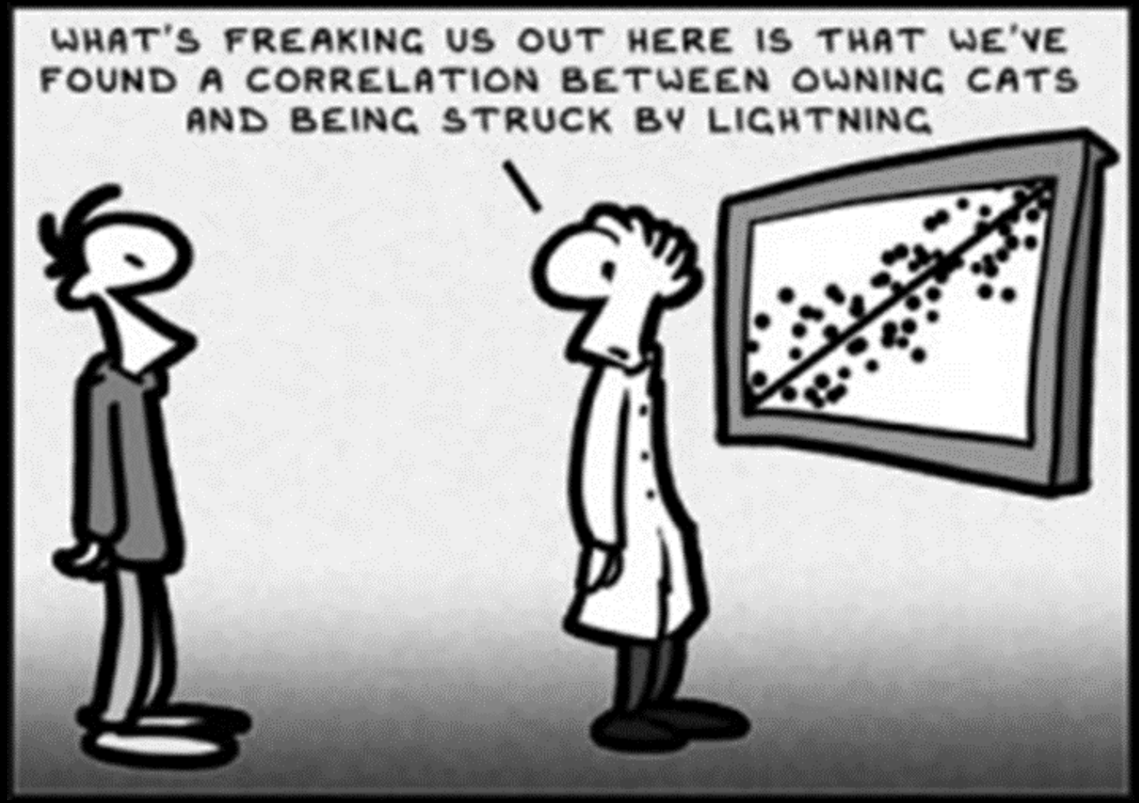
Positive products → positive covariance.

Correlation

- Relations among two or more variables
- How two variables relate each other?
- One variable may cause variation, or changes, in another variable

Principal drawback of correlational research

- Does not allow for the careful controls necessary for drawing conclusions about causal associations between variables.

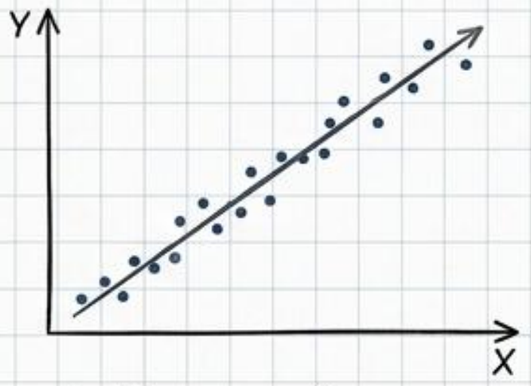


Correlation

- **Positive Correlation**
  - Values on the two variables move in the same direction
- **Negative Correlation**
  - On average, as scores on one variable go up, scores on the other variable go down
- **Strength or Magnitude of the relationship -1 to +1**
- **Correlation Coefficient**
  - Most commonly used is the **Pearson product-moment correlation coefficient**

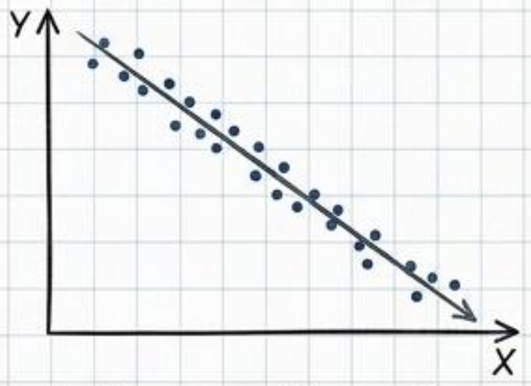
Correlation: Strength & Direction of Relationships

Strong Positive Correlation



As X goes up, Y goes up.
Points lie close to a line.

Strong Negative Correlation



As X goes up, Y goes down.
Points lie close to a line.

Pearson Correlation (r)

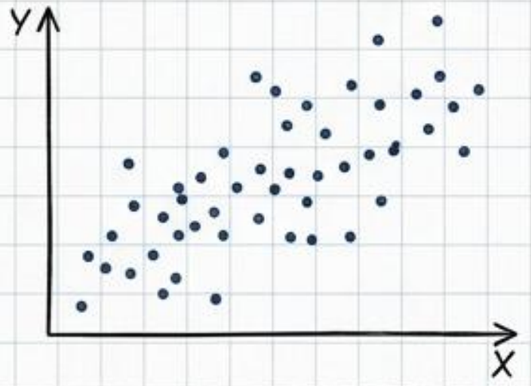
0
(No Relationship)

-1.0
(Perfect Negative)

+1.0
(Perfect Positive)

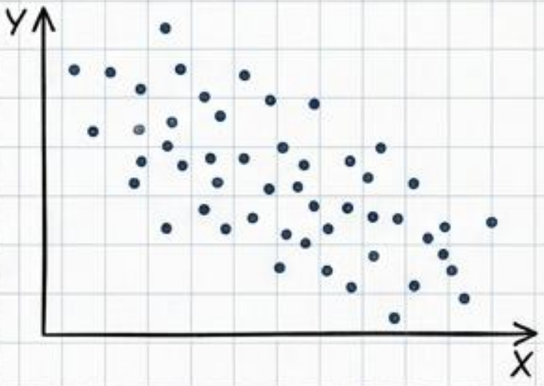
Measures the linear association between two variables.

Weak Positive Correlation



+0.3 to +0.7

Weak Negative Correlation



-0.3 to -0.7

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A Big Mistake!

- It is very unfortunate that people mistakenly make claims of causation as a function of correlations all the time.
  - Very common in advertisements and news stories

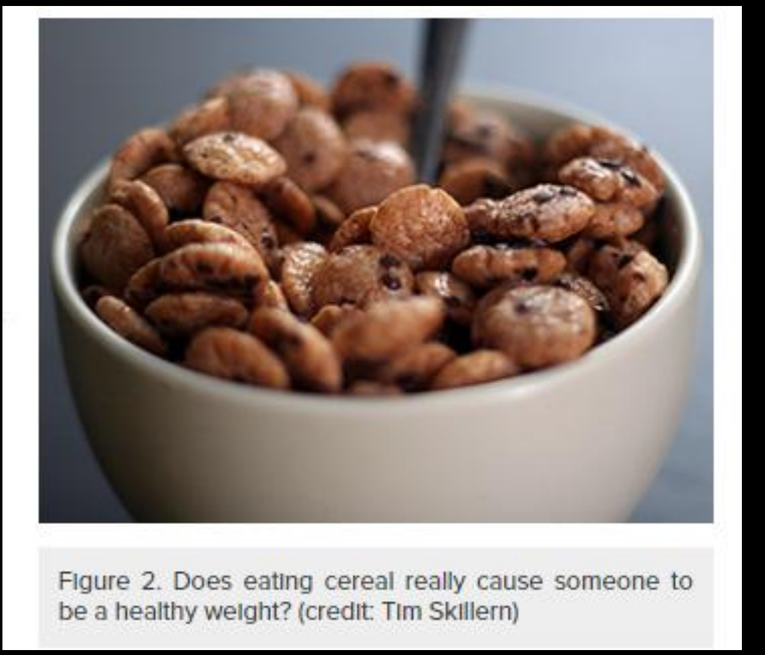
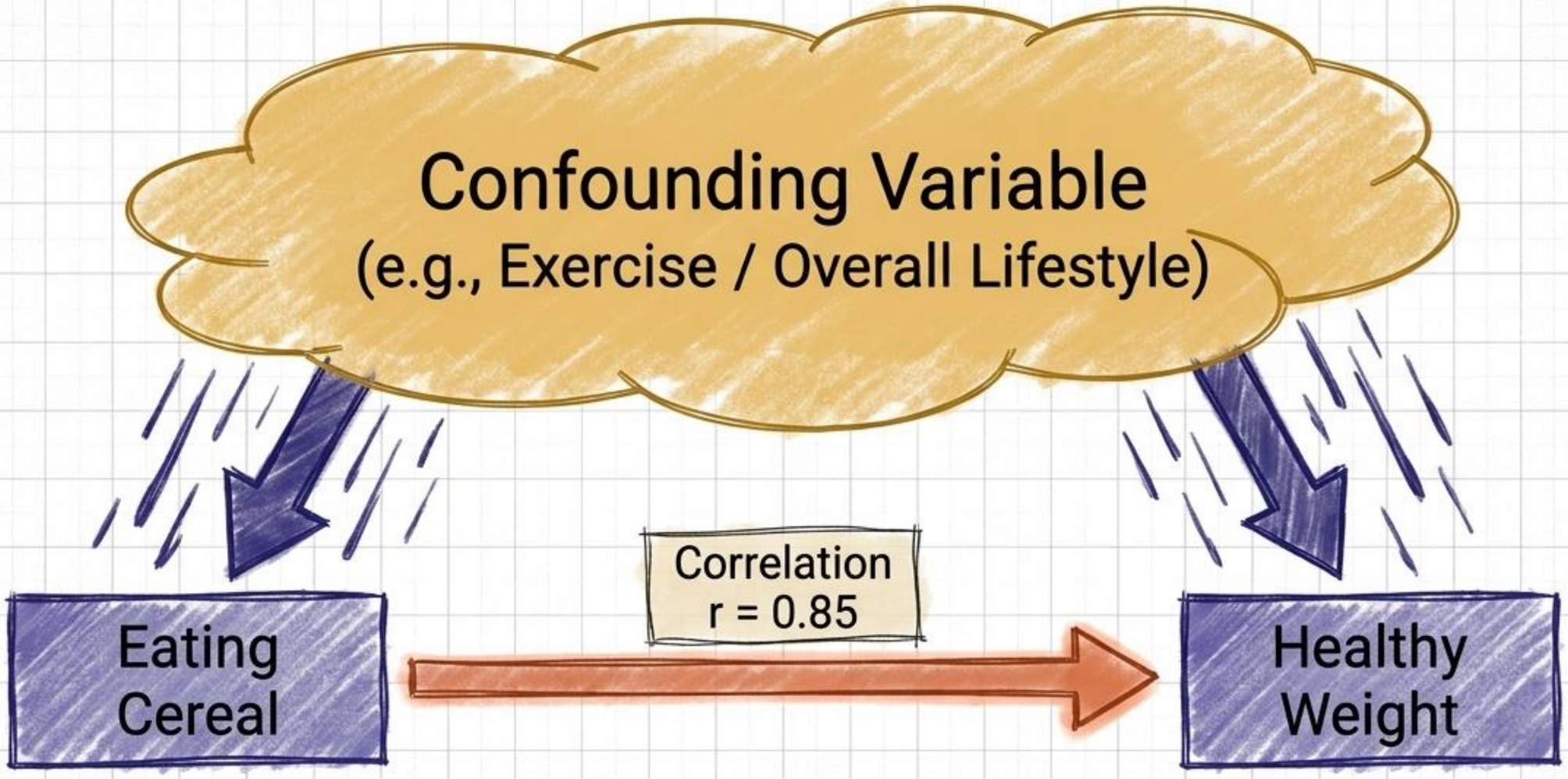


Figure 2. Does eating cereal really cause someone to be a healthy weight? (credit: Tim Skillern)

Correlation Does **Not** Imply Causation: The Confounding Variable



Evidence of a relationship between two variables is a necessary ingredient for causation, but it does not prove causation. A correlation only indicates that variables vary together.

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Correlation vs. Causation

- **Correlation**
  - variation in the scores on one variable corresponds with variation in the scores on a second variable
- **Causation**
  - variation in the scores on one variable ***causes or creates*** variation in the scores on a second variable

Evidence of a relationship between two variables (i.e., a correlation) does not necessarily mean that there is a causal relationship between the two variables

Pearson Product-Moment Correlation

$$r_{xy} = \frac{N\sum XY - (\sum X \sum Y)}{\sqrt{[N\sum X^2 - (\sum X)^2][N\sum Y^2 - (\sum Y)^2]}}$$

Covariance vs. Correlation

Covariance	Correlation
Shows **direction**	Shows **direction + strength**
Depends on units	Unitless
No fixed range	−1 to +1
Harder to compare	Easy to interpret

$$r = \frac{Cov(X,Y)}{S_X S_Y}$$

- S_X = standard deviation of X
- S_Y = standard deviation of Y

Simple takeaway:
Covariance: Do X and Y move together?
Correlation: How strongly do they move together?

Pearson Product-Moment Correlation

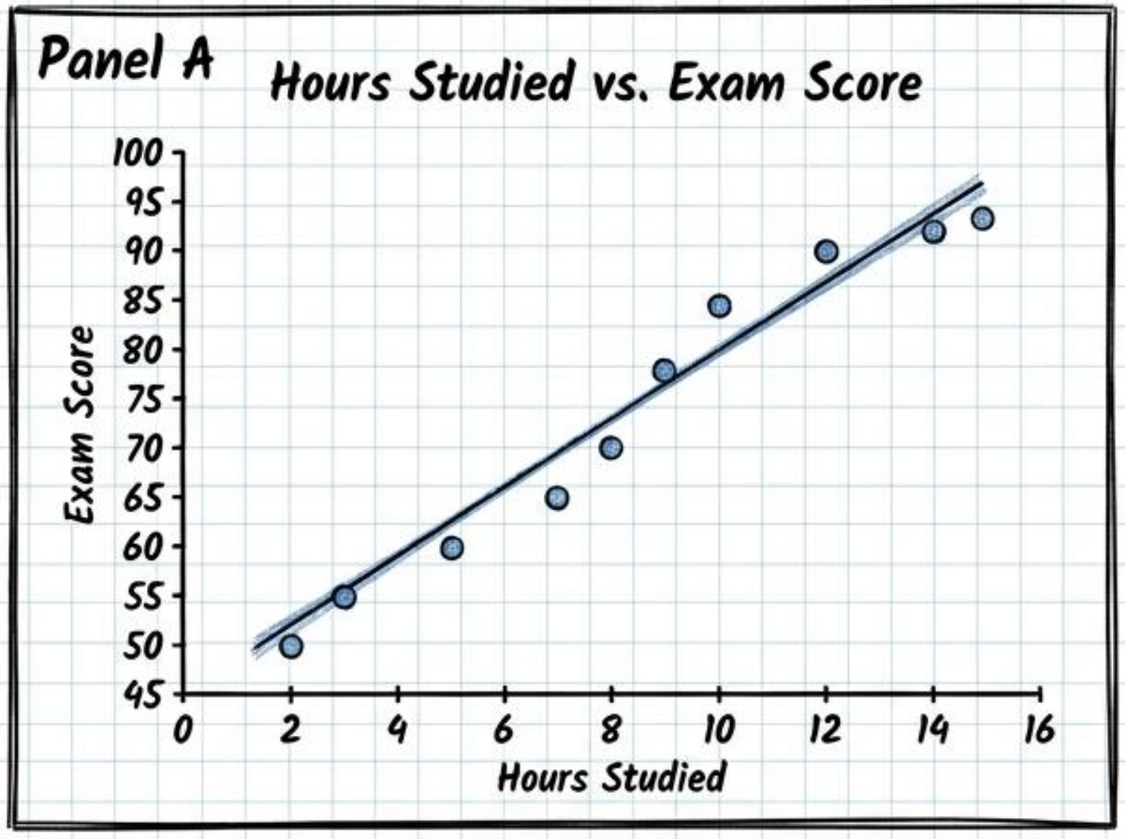
- Both of the variables must be
  - measured on an interval or ratio scale
  - continuous variables
- Simple Pearson correlations are designed to examine linear relations among variables

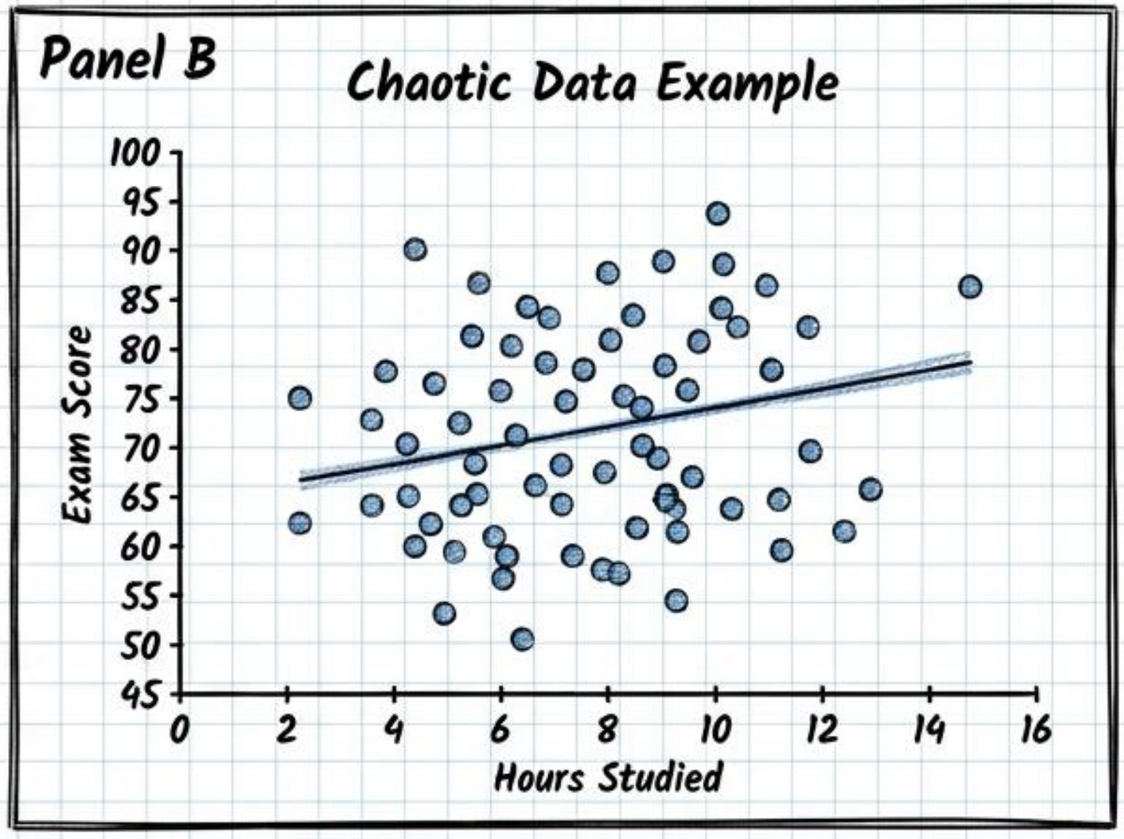
Coefficient of Determination

- Squared correlation coefficient is known as the ***coefficient of determination***
- Percentage of variance explained in each of the two correlated variables by the other variable

Coefficients of determination also do not necessarily indicate a causal relationship between the two variables

The Coefficient of Determination (R^2): Explained Variance





$R^2 = 0.96$

96% of the variability in exam scores is completely explained by the hours studied.

$R^2 = 0.10$

Only 10% of the mystery is explained. Other variables are causing the change.

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Regression

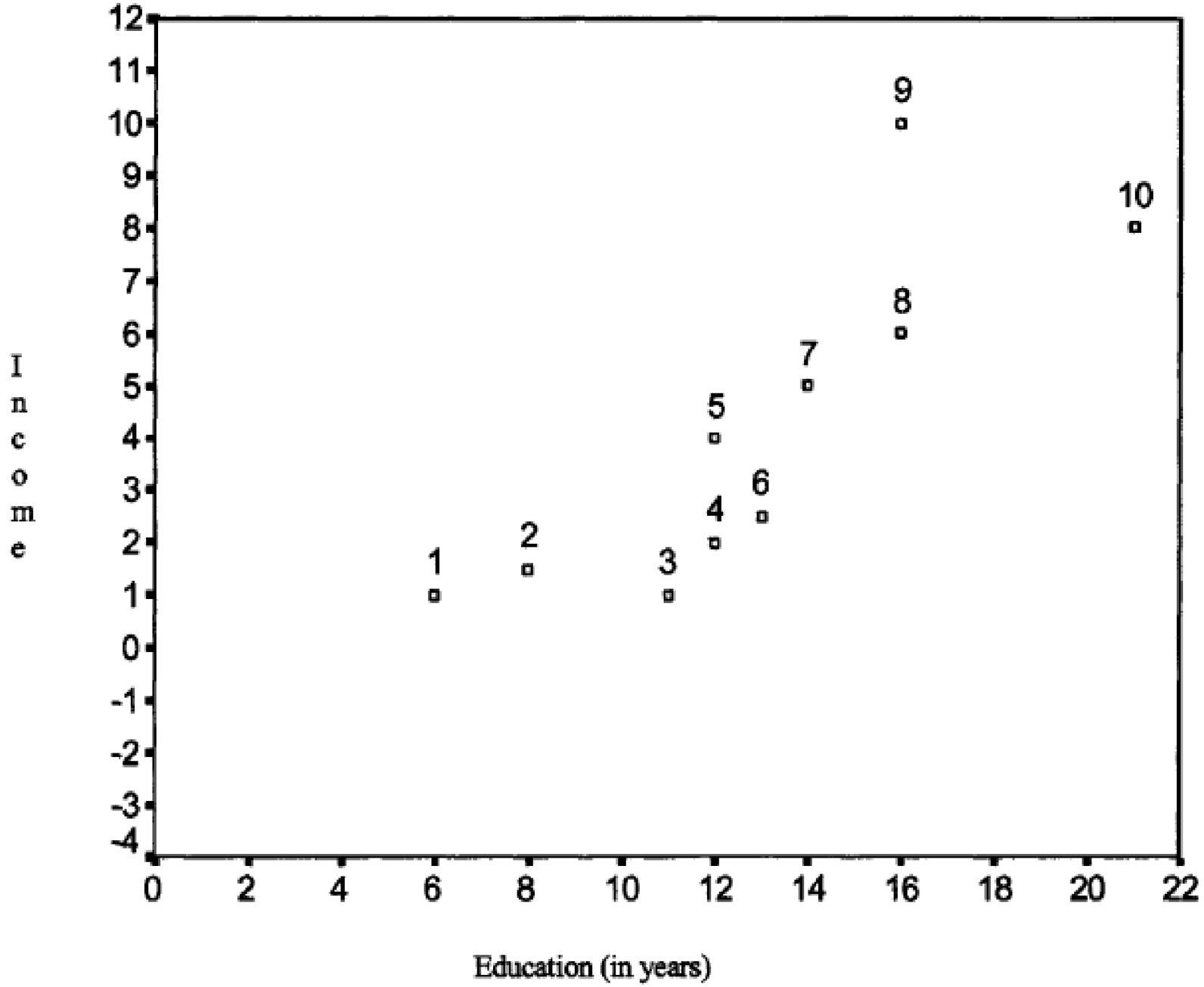
- Nature and strength of the relations between the variables.
- Purpose of regression analysis is to *make predictions* about the value of the *dependent variable* given certain values of the *predictor variable(s).*
- Stronger correlations will lead to more accurate predictions.
- Yields a formula for calculating the predicted value of one variable when we know the actual value of other variable(s).

$$\hat{Y} = bX + a$$

where $\hat{Y}$ is the predicted value of the Y variable
b is the unstandardized regression coefficient, or the slope
a is intercept (i.e., the point where the regression line intercepts the Y axis)

Linear Bivariant Regression

- Assumption: Two variables are ***linearly*** related
- Involves a single independent, or predictor variable and a single dependent, or outcome variable


Income
12
11
10
9
8
7
6
5
4
3
2
1
0
-1
-2
-3
-4
0
2
4
6
8
10
12
14
16
18
20
22
1
2
3
4
5
6
7
8
9
10
Education (in years)

Ordinary Least Squares Regression

- Minimizing the sum of squared deviations
- Line of least squares is the regression line
- This regression line represents the *predicted values* of Y at any given value of X.
- Calculate slope and intercept

Anatomy of an Equation: The Regression Line

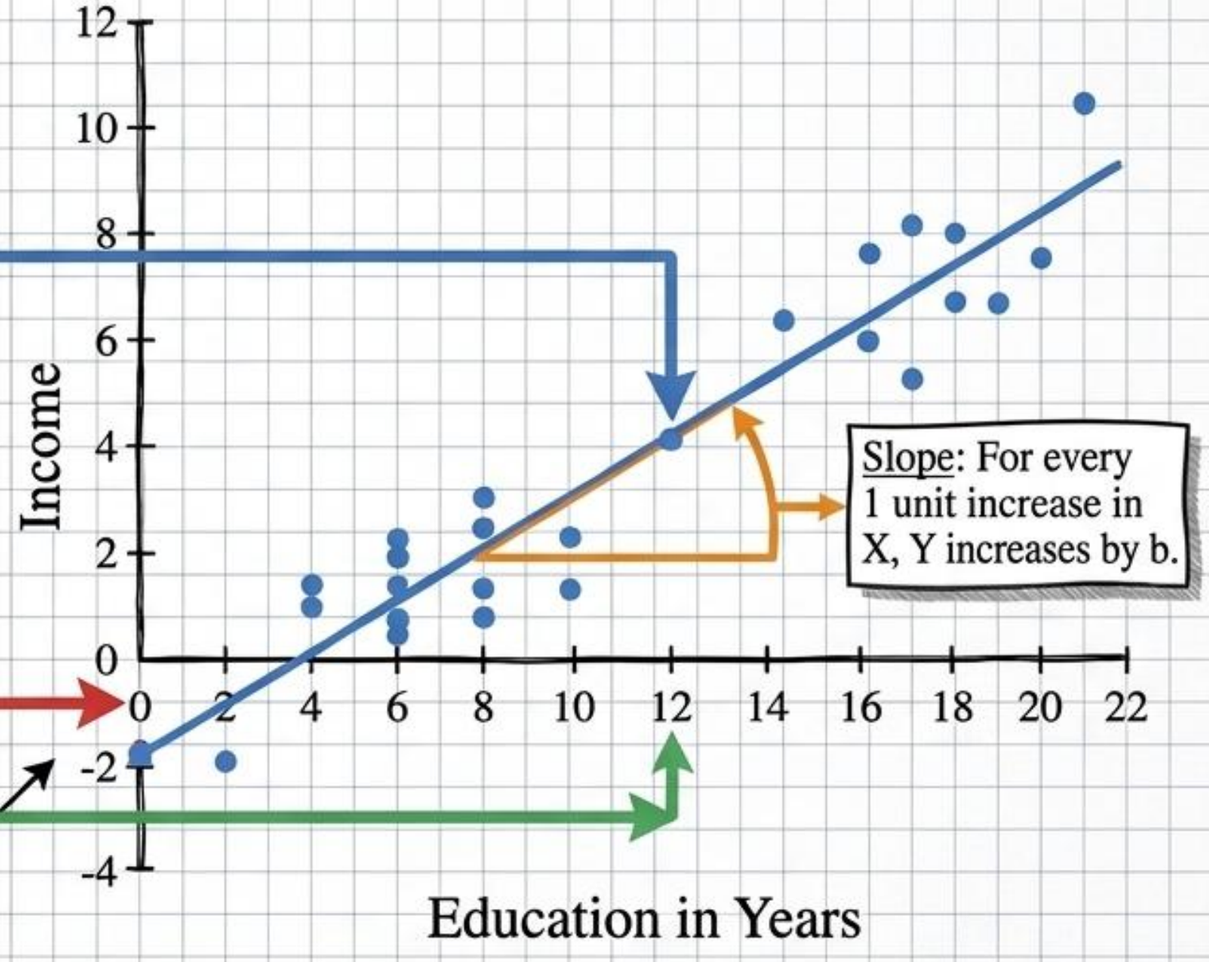
$$Y = bX + a$$

Predicted Value: The specific point on the line we guess for a given X.

Predictor: The independent variable.

Intercept: Exactly where the regression line crashes into the Y-axis when X=0.

Slope: For every 1 unit increase in X, Y increases by b.



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Regression Coefficient

- Or slope of regression line

$$b = \frac{\sum(x - \bar{x}) * (y - \bar{y})}{\sum(x - \bar{x})^2}$$

Intercept of Regression Line

a =intercept

$$a = \bar{Y} - b\,\bar{X}$$

where $\bar{Y}$ is the average value of Y,
$\bar{X}$ is the average value of X,
and b is the regression coefficient.

The Coefficient of Determination

Understanding R^2 in Regression Analysis

Definition:

R^2, or the coefficient of determination, is a statistical measure that explains the proportion of variance in the dependent variable that is predictable from the independent variable(s).

$$R^2 = 1 - \frac{\sum(Y_{\text{actual}} - Y_{\text{predicted}})^2}{\sum(Y_{\text{actual}} - Y_{\text{mean}})^2}$$

Visualizing R^2

Scatter Plot Example:

- Show a scatter plot with the regression line.
- **High R^2:** Points close to the regression line.
- **Low R^2:** Points scattered far from the regression line.

Interpretation:

- If $R^2=1$, the regression line perfectly fits the data.
- If $R^2=0$, the independent variable explains none of the variability in the dependent variable.

HW#1

Here's a practice question for your students on bivariate regression:

Question: A researcher wants to study the relationship between the number of hours students spend studying and their final exam scores. The following data shows the number of hours studied and the corresponding final exam scores for 10 students:

Student	Hours Studied (X)	Exam Score (Y)
1	2	50
2	3	55
3	5	60
4	7	65
5	8	70
6	9	78
7	10	85
8	12	90
9	14	92
10	15	95

Using the data in the image:

- **X = Hours Studied**
- **Y = Exam Score**
- **n = 10**

We want the **bivariate (simple linear) regression equation**:

$$Y = a + bX$$

1. Calculate the means

$$\bar{X} = \frac{85}{10} = 8.5$$

$$\bar{Y} = \frac{740}{10} = 74$$

2. Calculate the regression coefficients

The slope is:

$$b = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sum(X - \bar{X})^2}$$

For these data:

$$b = 3.63897$$

The intercept is:

$$a = \bar{Y} - b\bar{X}$$

$$a = 74 - (3.63897)(8.5)$$

$$a = 43.0688$$

Therefore, the regression equation is:

$$\boxed{\hat{Y} = 43.069 + 3.639X}$$

Interpretation

Slope = 3.639:

For every additional **1 hour of studying**, the predicted exam score increases by approximately **3.64 marks**.

Intercept = 43.069:

The model predicts a score of about **43.1** when study time is zero. This is the mathematical intercept; it doesn't necessarily have a practical interpretation.

Goodness of fit

The correlation coefficient is:

$$r = 0.9837$$

and

$$R^2 = 0.9676$$

So approximately **96.8% of the variation in exam scores is explained by hours studied** in this small dataset.

For example, for a student studying **10 hours**:

$$\hat{Y} = 43.069 + 3.639(10)$$

$$\boxed{\hat{Y} \approx 79.46}$$

So the regression model predicts an exam score of approximately **79.5 marks** for 10 hours of study.

Predicted Value vs. Actual Value of Y

- In making predictions about the value of Y, we may be incorrect sometime

$$e = Y - \hat{Y}$$

$$Y = bX + a + e.$$

- Errors in prediction are known as **residuals**



The Residual (Error)

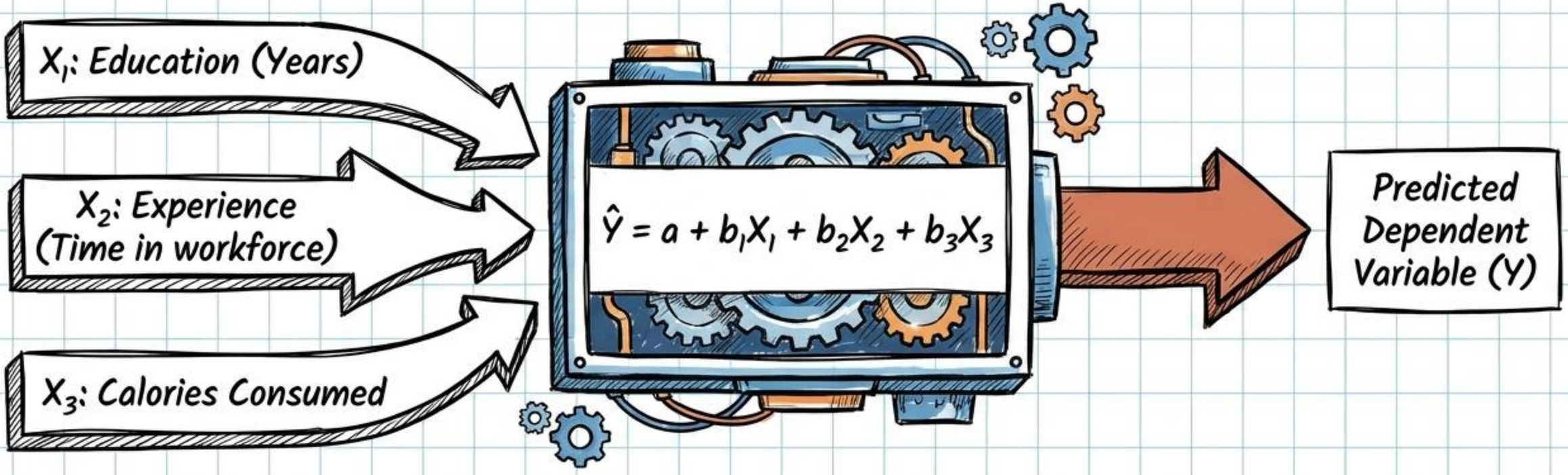
$e = Y - \hat{Y}$ (Actual Value minus Predicted Value).

Ordinary Least Squares (OLS)

The statistical engine behind regression. The regression line is drawn exactly where it minimizes the total sum of all squared residual gaps across the entire dataset. It is the path of least error.

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Multiple Regression: The Multivariate Engine



Core Insight: Multiple regression tests the unique predictive power of each independent variable while controlling for the overlapping effects of all the others.

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The Statistical Toolbox		
Your Goal	**The Variables You Have**	**The Tool to Use**
Summarize a single variable's distribution.	Continuous.	Mean, Median, Mode, Standard Deviation.
Find the strength and direction of a relationship.	Two Continuous Variables.	Pearson Correlation (r).
Predict an outcome based on a single input.	One Predictor, One Dependent (Interval/Ratio).	Simple Linear Regression (OLS).
Predict an outcome based multiple real-world factors, controlling for overlap.	Multiple Predictors, One Dependent.	Multiple Regression.

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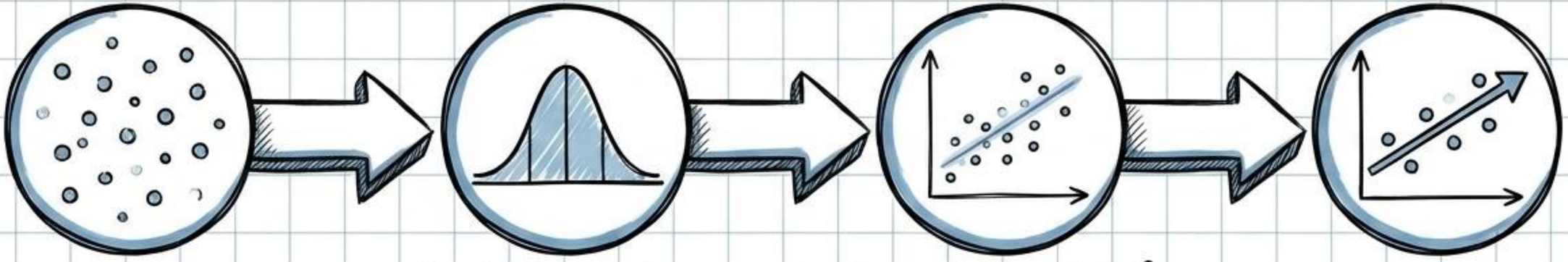
The Statistical Blueprint: From Data to Insight

Data Collection & Sampling

Central Tendency & Spread

Correlation & R^2

Predictive Models



Central Tendency & Spread

Correlation & R^2

Statistics never lie, but interpretations can. You now possess the blueprint to describe the present, infer the unknown, and predict the future. Interrogate your data responsibly.

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